Anomaly Detection Algorithm for Enhanced Fraud Identification in the Insurance Sector

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Abstract

Detecting fraud in the vast databases of the insurance industry remains an ongoing challenge. Existing algorithms often flag numerous suspicious cases, resulting in time-consuming and costly investigations. To address this issue, we propose a novel, sequential, and multilayered algorithm that achieves a high level of accuracy while presenting a manageable number of cases for expert review. In this article, we evaluate the performance of our algorithm using real-world data, demonstrating its impressive accuracy of over 95% and notable speed. This approach offers a significant advancement in fraud detection for the insurance sector.

Keywords: Fraud Detection; Supervised/Unsupervised/Reinforcement Learning; Artificial Intelligence; Insurance Application; Multilayer Algorithms.

MSC2010: 62D05, 62F10, 62N01, 65C05.

1 Introduction

Fraud poses a significant ethical and financial threat to the global insurance industry. Committed by both policyholders and, at times, insurers themselves, fraudulent activities can have severe consequences. As high levels of fraudulent claims lead to substantial financial losses, insurance companies must take proactive steps to prevent fraud to avoid increasing premiums for all customers. Even a small percentage of dishonest policyholders can result in higher premiums across the board. In 2019, the European Insurance and Reinsurance Federation reported an alarming 13 billion euros in

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industry fraud, underscoring the urgent need for efficient and effective fraud prevention measures [Hilal et al., 2022].

The inherent stochasticity of insurance claim losses makes identifying suspicious and fraudulent cases a complex and time-consuming task, often demanding significant resources. In recent years, machine learning techniques have proven to be powerful tools in detecting insurance fraud. Both supervised and unsupervised learning approaches have demonstrated considerable potential in enhancing the accuracy and efficiency of fraud detection systems [Hilal et al., 2022].

To address the multifaceted challenges posed by fraud, this research concentrates on enhancing fraud detection methodologies. Our goal is to develop more efficient and accurate techniques that reduce the number of flagged cases while maintaining a high detection rate. By doing so, we aim to alleviate the burden on insurance companies and their experts, enabling them to concentrate their efforts on the most relevant and pressing cases.

In this study, we introduce a novel sequential and multilayered anomaly detection algorithm designed to strike a balance between accuracy and efficiency. Our approach not only contributes to minimizing financial losses resulting from fraudulent activities but also helps restore trust between insurance companies and their customers. Moreover, by improving the overall fraud detection process, businesses can potentially reduce the substantial annual losses they currently face. Also, we seek to provide practical solutions that can be readily implemented in the insurance industry, ultimately benefiting both insurance providers and policyholders. By streamlining the fraud detection process and mitigating the adverse effects of fraudulent activities, we strive to create a more resilient and trustworthy insurance landscape.

In this article, we present an innovative approach that combines advanced machine learning techniques, such as reinforcement learning and supervised learning, to create a highly accurate and automated fraud detection system. Our goal is to surpass traditional manual methods by reducing the reliance on human intervention, accelerating detection speed, and enhancing the overall efficiency of fraud prevention measures.

By leveraging these cutting-edge methods, we aim to provide insurance companies with a powerful tool to combat fraudulent activities. Additionally, we offer an overview of various existing techniques employed in detecting fraud, providing a comprehensive understanding of the current landscape and the potential advantages of our proposed approach.

The rest of this paper is organized as the following. Section 2 provides a

literature review on fraud detection in the insurance industry. We describe our proposed algorithm in Section 3. In Section 4, we implement the algorithm using real-world data and present evaluation metrics. Concluding remarks and discussions are given in Section 5.

2 LITERATURE REVIEW

Insurance fraud detection is a critical task for data science and machine learning. Given the inherent complexities of the problem, the application of various machine learning techniques for fraud detection in the insurance industry has gained significant attention.

Han et al. (2011) stressed the importance of clustering, classification, and outlier analysis as crucial methods in their work, serving as a foundational reference for understanding core data mining challenges [Han et al., 2011]. Ngai et al. (2011) investigated data mining techniques in financial fraud detection, presenting a classification framework and an extensive literature review that compiles key methods and applications [Ngai et al., 2011].

Aggarwal et al. (2015) explored the various applications of data mining techniques in financial fraud detection, demonstrating how these methods can substantially improve fraud detection accuracy [Aggarwal, 2015]. Aggarwal and Philip (2015) offered a thorough review of AI techniques in fraud detection, emphasizing the role of machine learning methods in advancing fraud detection systems [Aggarwal and Philip, 2015].

In a recent study, Yao et al. (2019) applied the Hampel Filter to eliminate profile-isolated outliers in laser vision measurement, highlighting the importance of outlier detection in enhancing measurement accuracy [Moon et al., 2019]. Moreover, Moon et al. (2019) employed predictive modeling for identifying fraudulent automobile insurance claims, leveraging advanced machine learning algorithms to enhance fraud detection effectiveness [Moon et al., 2019]. These studies emphasize the ongoing advancements in outlier detection and predictive modeling techniques, demonstrating their potential in addressing fraud-related challenges in various domains, including insurance.

Bryant and Desai (2020) further expanded on this discussion by focusing on the practical applications of machine learning in financial fraud detection, emphasizing the pivotal role these technologies play in modern fraud prevention strategies [Bryant and Desai, 2020]. More recently, Kou et al. (2021) explored AI-based fraud detection in the insurance and financial industries,

offering valuable insights into the integration of AI in fraud detection systems to bolster efficiency and accuracy [Kou et al., 2021]. These studies highlight the increasing importance of machine learning and AI in the realm of fraud detection, showcasing their potential to revolutionize traditional fraud prevention approaches and enhance overall system performance.

In a recent study, Zhang et al. (2022) delved into the development of auto insurance knowledge graphs and their application in fraud detection, exhibiting substantial progress in leveraging knowledge graphs for practical fraud identification purposes [Zhang et al., 2022]. Their work contributes to the ongoing advancements in data-driven fraud detection techniques, emphasizing the importance of integrating domain-specific knowledge for enhanced performance.

In a recent contribution, Nijhuis and van Lelyveld (2023) investigated the application of reinforcement learning for outlier detection in costly-to-verify data, proposing novel methods to tackle the challenges associated with outlier handling [Nijhuis and van Lelyveld, 2023]. Their work signifies the importance of exploring advanced machine learning techniques to overcome practical constraints and improve the accuracy of fraud detection systems.

Collectively, these studies underscore the potential of employing diverse machine learning and AI techniques to enhance insurance fraud detection, augmenting the precision and efficiency of detection systems. The research emphasizes the importance of continued exploration and development of advanced methodologies to address the challenges posed by insurance fraud effectively.

3 METHODOLOGY

Our proposed algorithm consists of several key components: Data Preprocessing, Anomaly Detection, Machine Learning, Model Evaluation, and Structures. In the following subsections, we provide a detailed description of each part within the algorithm to highlight the unique features and functionalities of our approach.

3.1 PRE PROCESSING

Dimensionality reduction is a crucial preprocessing technique in machine learning that focuses on decreasing the number of input variables, or features,

in a dataset. This process involves projecting data from a high-dimensional space into a lower-dimensional space while preserving the significant characteristics of the original data. The following section will introduce two popular and broadly applicable methods for data preprocessing that effectively employ dimensionality reduction.

Dimension Reduction

In this research, we employ the two following methods for dimensionality reduction and subsequently partition the dataset randomly into m subsets. Further details on this process can be found in Algorithm 1.

Gini Index: The Gini index measures inequality in a dataset, ranging from 0 (all observations are equal) to 1 (all observations are different). It's calculated by $Gini(Y) = 1 - \sum_{r=1}^{R} P_r^2$, where the lower values indicate the more equal distribution [Giudici et al., 2024].

Near Zero Variance: The Near Zero Var function identifies predictors with limited variability, such as single-value or low-variance predictors, and those with a high frequency ratio between the most common and second most common values [Giudici et al., 2024].

Algorithm 1 PreProcessing Algorithm

inputs : Dataset D , Partitioning percentages $p_i\,, i = 1, \cdots, m$,
bounds for Gini Index $g_L\,, g_U$,
outputs: Partitions D_i, $i = 1, \cdots, m$,
Remove The Categorical columns of D where Gini Index is not between $g_L\,, g_U$,
Remove The Numerical columns of D where Near, Zero Variance is TRUE,
Separate D into m Partitions where the size, of D_i is proportional to $p_i\,, i = 1, \cdots, m$.

Balancing

Resampling structures:
The Synthetic Minority Over-sampling Technique (SMOTE,) is an approach used to balance imbalanced datasets by generating synthetic samples from the minority class. SMOTE operates by randomly selecting a sample from the minority class and identifying one of its k-nearest neighbors. A synthetic sample is then generated through interpolation between these two samples. This technique enhances the performance of machine learning models when

dealing with imbalanced data [Fajardo et al., 2021]. This study incorporate the SMOTE method for enhancing machine learning algorithms in fraud detection in insurance industry.

Structure

We now introduce a schematic representation of two distinct structures within our proposed Sequential Anomaly Detection (SAD) algorithm, designed specifically for fraud identification in the insurance industry. These structures form the basis of our innovative approach to tackling fraudulent activities within this domain.

Figure 1 illustrates a sequential process involving several stages: First, input data undergoes preprocessing to reduce its size and is subsequently divided into desired subsets. Anomaly detection is then applied to the first subset, with the identified potential fraud cases sent to an expert for verification. This step also enables fraud pattern recognition and the development of a trained model using machine learning algorithms. In subsequent iterations, the trained model is employed to detect potential fraud data, allowing for expert confirmation and continuous refinement of the machine learning algorithms to achieve a more accurate fraud detection system.
Figure 2 illustrates a multilayered structure where the EAD method is applied to the result of anomaly detection to get the more suspicious cases as the potential frauds.

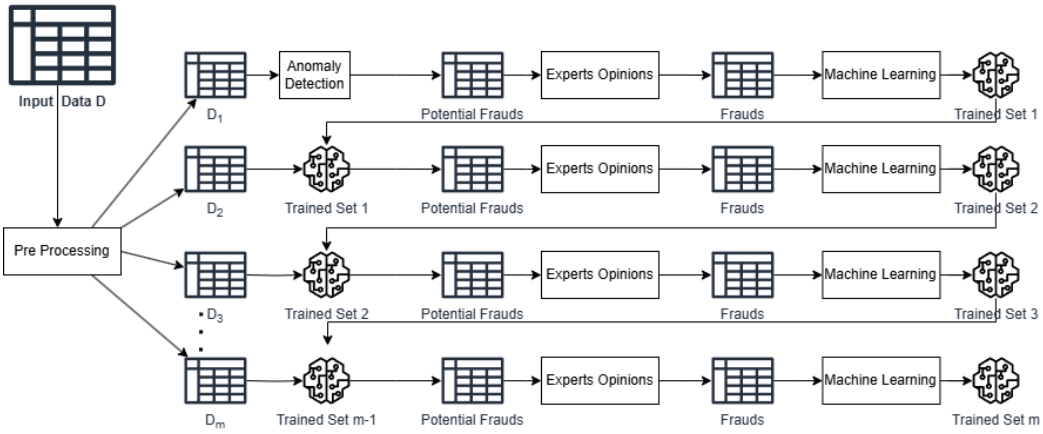


Figure 1: Workflow of simple structure of sequential anomaly detection. (SAD)

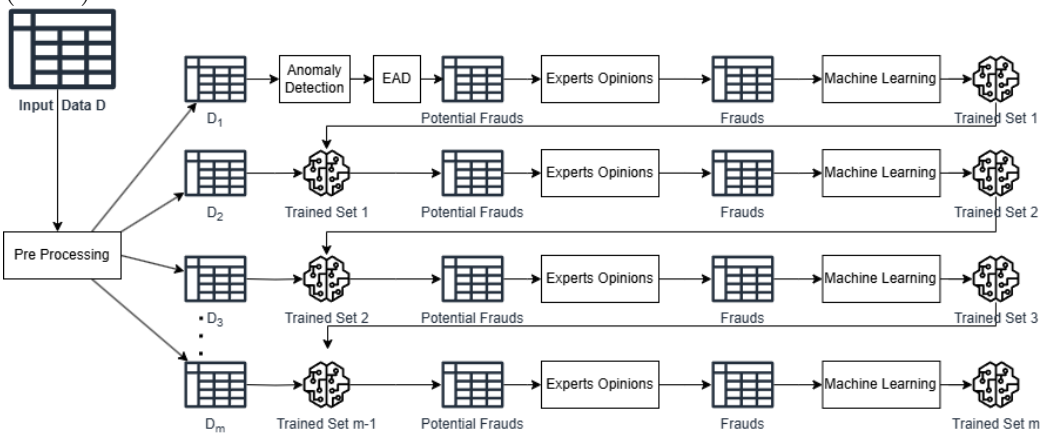


Figure 2: Workflow of multilayered structure of sequential anomaly detection.(SAD)

ANOMALY DETECTION

In the initial stage of the algorithm, the absence of reliable labels for fraud or suspicious data necessitates unsupervised models and techniques to identify potential fraudulent activities. Various anomaly detection algorithms have

been proposed by researchers, and the models employed in our study are introduced below.

Box Plot: Box plots aid in identifying outliers by examining values that fall outside the range of $Q3 + (1.5 \times IQR)$ or $Q1 - (1.5 \times IQR)$, where $Q1$ and $Q3$ represent the first and third quartiles, respectively, and IQR is the interquartile range. Extreme points are defined as those beyond $Q3 + (3 \times IQR)$ or $Q1 - (3 \times IQR)$ [Hanafy and Ming, 2021].

Percentile structure: Percentiles partition data into 100 segments, indicating the percentage of data points that fall below a specific value. The 25_{th} percentile, for instance, marks the point where 25% of the data lies below it. Outliers typically fall below the 5_{th} percentile or above the 95_{th} percentile, meaning that normal data usually resides within the $(5, 95)\%$ range [Aggarwal et al., 2019].

Z Score structure: Z Score identifies outliers by comparing individual data points to the mean and standard deviation of the dataset. Values with a Z Score above 3 or below -3 are considered outliers. This straightforward method proves effective when dealing with data exhibiting a normal distribution [Aggarwal et al., 2019].

Hample filter structure: The Hampel Identifier detects outliers using the median and median absolute deviation. For data sequences with a moving window, a point is an outlier if its difference from the local median exceeds a certain threshold. Usually, the threshold is set to 3. If set to 0, it acts like a standard median filter [Yao et al., 2019].

E Grubs test structure: The Grubb's test detects outliers in normally distributed data by contrasting the test statistic against a critical value. It operates under the assumption that the data contains either no outliers or a single outlier. This test is brief and easy to implement [Nguyen et al., 2019].

KNN structure: The K-Nearest Neighbors (KNN) algorithm calculates the distance between a test instance and training examples to identify its closest neighbors. While this process can be computationally intensive for large datasets, various indexing techniques can help improve its efficiency [Sikder and Batarseh, 2023].

Local outlier factor (LOF) structure: Local Outlier Factor (LOF) detects outliers by employing KNN to compare each data point with its neighbors, evaluating the isolation of a point based on its local reachability density. A simplified LOF approach, while more computationally demanding, utilizes KNN distance to facilitate density estimation, resulting in improved performance [Sikder and Batarseh, 2023].

Isolation forest structure: The Isolation Forest (iForest) algorithm efficiently detects outliers by constructing binary trees from random dataset partitions, calculating anomaly scores for each data point. It exhibits linear time complexity and high precision [Tan et al., 2016]

K Modes structure: The K Modes clustering algorithm, an alternative to K-means, efficiently clusters categorical data by iteratively selecting k centroids, calculating dissimilarity, assigning objects to the nearest centroid, and updating centroids. This straightforward approach ensures optimal clustering performance [**?**].

Entropy Anomaly Detection (EAD): Anomaly detection using information theory calculates a dataset's information content and observes changes when removing potential anomalies. Higher information gain, defined as $\text{Gain}(x) = \text{Info}(D) - \text{Info}(D \backslash x)$, indicates an anomaly. Here, the information content of dataset D is given by $\text{Info}(D) = \sum_{x \in D} p(x) \times \log(p(x))$ [Subudhi and Panigrahi, 2020].

Identifying anomalies can be achieved by eliminating instances with the highest Gain(X). Efficient approximations are employed to address the computational complexity associated with this process.

Threshold limits: In anomaly detection, it is essential to introduce suspicious cases identified by most methods. This is achieved using the anomaly detection threshold limit, denoted as T_{AD}, which determines the criteria for classifying data points as potential anomalies.

3.2 MACHINE LEARNING

Following anomaly detection, the data is labeled with expert opinions, enabling the application of supervised learning. Our supervised learning algorithm utilizes various machine learning techniques, which are outlined in subsequent sections.

LR models: Logistic Regression (LR) is a popular supervised learning model used to predict outcomes based on input features and target variables [Tan et al., 2016].

SVM models: The Support Vector Machine (SVM) is a widely used machine learning classifier in various real-world applications, addressing both linear and nonlinear problems. It separates instances of classes using a hyperplane, making it ideal for nonlinear classification and complex problems [Tan et al., 2016].

XGB models: XGBoost is an efficient gradient boosting implementation that builds additive models using decision trees. It creates an ensemble of base learners by iteratively minimizing errors, outperforming individual classifiers. [Subudhi and Panigrahi, 2020].

RF models: Breiman in 2001 introduced the Random Forest (RF) classifier which combines bagging and subspaces to construct a set of decision trees from the training dataset. Predictions are made based on votes from these trees, yielding high accuracy. [Tan et al., 2016].

NB models: Naïve Bayes (NB) is an efficient predictive model for large datasets, utilizing Bayes' theorem and assuming predictor independence. In fraud detection, the model calculates probabilities with Maximum A Posteriori (MAP) and is distinct due to its feature distribution assumptions [Tan et al., 2016].

ADAB Boost models: Adaptive Boosting, or Adaboost, is a renowned machine learning algorithm known for its quick boosting capabilities. Boosting algorithms transform lazy learners into eager learners, enhancing their predictive abilities through training[Tan et al., 2016].

ANN models: Artificial Neural Networks (ANNs) are models inspired by the human brain, capable of learning patterns from data and making predictions. ANNs comprise interconnected artificial neurons processing information through layers. By adjusting weights and biases, they adapt to complex data relationships [Tan et al., 2016].

Threshold limit: In machine learning methods, it is crucial to identify cases that most algorithms have flagged as suspicious under the potential frauds category. This is accomplished using the machine learning threshold limit, denoted as T_{ML}, which establishes the criteria for classifying data points as potential fraud cases.

3.3 Sequential Anomaly Detection (SAD)

In following, we describe two structures where the input is preprocessed data in Algorithm 1.

Simple structure of the SAD algorithm

The anomaly detection process starts with unlabeled data from the initial section. After voting-based labeling, it predicts new instances using machine

learning methods (supervised learning). At the end of each section, models are updated with expert opinions, enabling continuous improvement in anomaly detection.

The structure takes multiple inputs, including the number of machine learning models, anomaly detection methods, minimum votes for machine learning models and anomaly detection methods, as well as the set of machine learning models and anomaly detection methods. Additionally, the dataset provided to Algorithm 1 is a critical input for the structure. These inputs contribute to a comprehensive preprocessing phase in the anomaly detection process.

In the initial step, all data from the first section is processed by anomaly detection methods. Each observation and method is labeled as either normal (assigned a value of 0) or anomalous (assigned a value of 1) based on the method's determination. Next, for each observation in the first section, the labels received from each method are summed, referred to as the total votes for that observation, U_{Total}. If U_{Total} is greater than or equal to the threshold T_{AD}, the observation is flagged as suspicious; otherwise, it is considered normal.

The set F_1 consists of all observations identified as suspicious and is submitted to experts for further analysis. After review, each observation in F_1 receives a new label (0 or 1), indicating whether fraud is present, forming the set E_1. Since E_1 is likely to have imbalanced labels, balancing methods are applied to make it suitable for use in machine learning models. To achieve better balance, PN percent of normal data from the first section is combined with the balanced E_1, creating the new balanced set E_1'.

For each machine learning model, the balanced set E_1' is used for training, and then all observations in the second data section are classified using each trained model. Every model contributes a vote for each observation, and if the total votes exceed the threshold T_{ML}, the observation is flagged as an outlier for expert review. Data balancing and model updating are performed iteratively for subsequent sections, ensuring that the models remain current throughout the entire dataset analysis.

Algorithm 2 Simple structure of the SAD algorithm

inputs : Dataset D , AD SET , ML SET , Expert opinion , PN
outputs : trained models , potential frauds
$N \leftarrow$ number of anomaly detection methods.
$M \leftarrow$ number of machine learning models.
AD SET $\leftarrow (AD_1, \cdots, AD_N)$
ML SET $\leftarrow (ML_1, \cdots, ML_M)$
$T_{AD} \leftarrow$ Threshold number of voting for anomaly detection methods
$T_{ML} \leftarrow$ Threshold number of voting for machine learning models
PreProcessing on the input dataset D which is described in Algorithm 1
Store the result of anomaly detection methods of AD SET on D_1 into $U_1, \cdots, U_N$ (0 for normal & 1 for anomaly).
for Each row of D_1 **do**
    Set $U_{Total} = \sum_{i=1}^{N} U_i$
    if $U_{Total} \geq T_{AD}$ **then**
        Label the row as Potential Fraud
    else
        Label the row as Potential Normal
    end if
end for
Create F_1 dataset as the subset of D_1 for which labeled as Potential Fraud.
Send F_1 dataset for experts.
Receive the opinion of experts on F_1 dataset as E_1
Create E_1' as oversampling SMOTE model of $E_1 \cup G_1$ where G_1 is a random subset of size PN% of D_1 for which labeled as Potential Normal
for each Model of ML SET **do**
    Learn the Model with E_1' Dataset as training set.
end for
Set trained models as TS_1
for j from 2 to m **do**
    for each row of D_j **do**
        Use TS_{j-1} for prediction on D_j as tast set.
        Store the response prediction of TS_{j-1} on D_j into $V_1, \cdots, V_M$ (0 for normal & 1 for anomaly).
        Set $V_{Total} = \sum_{i=1}^{M} V_i$
        if $V_{Total} \geq T_{ML}$ **then**
            Label the row as Potential Fraud
        else
            Label the row as Potential Normal
        end if
        Create F_j dataset as the subset of D_j for which labeled as Potential Fraud.
        Send F_j dataset for experts.
        Receive the opinion of experts on F_j dataset as E_j
        Create E_j' as oversampling SMOTE model of $E_j \cup G_j \cup E_{j-1}'$ where G_j is a random subset of size PN% of D_j for which labeled as Potential Normal
        for each Models of ML SET **do**
            Learn the Model with E_j' Dataset as training set.
        end for
        Set trained models as TS_j
    end for
end for

Multilayered structure of the SAD algorithm

The Multilayered structure of the SAD algorithm processes data from the preprocessing algorithm 1, identifying anomalies in the unlabeled first section data through voting-based labeling. It then predicts new instances in subsequent sections using machine learning methods and updates models with expert feedback at the end of each section.

The structure's inputs include: the number of machine learning models, anomaly detection methods, minimum votes for machine learning models and anomaly detection methods, the set of machine learning models and anomaly detection methods, and the dataset provided to the preprocessing algorithm 1.

In the first step, all first-section data is processed by anomaly detection methods. Each observation and method is labeled as normal (0) or anomalous (1) based on the method's assessment.

In the subsequent step, the sum of labels from each method is calculated for every observation in the first section, denoted as U_{Total}. If U_{Total} exceeds the threshold T_{AD}, the observation is flagged as suspicious; otherwise, it is considered normal.

The set Y_1 comprises all observations suspected of fraud. These observations undergo re-examination using the entropy method EAD. If EAD also identifies them as suspicious, they are included in the dataset F_1 for expert review. After expert analysis, each observation receives a new label indicating fraudulence (0 or 1), forming the set E_1.

The set E_1 likely contains imbalanced labels, necessitating balancing for effective use in machine learning models. To achieve better balance, PN percent of normal data from the first section is combined with E_1, yielding the balanced set E_1'. This balanced set can then be utilized for training machine learning models.

Each machine learning model is trained on the balanced dataset E_1' and then classifies observations in the second data section. If the total votes for an observation surpass T_{ML}, it is flagged as an outlier for expert review. Datasets are balanced and models updated iteratively for subsequent sections, maintaining optimal performance and up-to-date models throughout the process.

Algorithm 3 Multilayered structure of the SAD algorithm

inputs : Dataset D , AD SET , ML SET , Expert opinion , PN
outputs : trained models , potential frauds
$N \leftarrow$ number of anomaly detection methods.
$M \leftarrow$ number of machine learning models.
AD SET $\leftarrow (AD_1, \cdots, AD_N)$
ML SET $\leftarrow (ML_1, \cdots, ML_M)$
$T_{AD} \leftarrow$ Threshold number of voting for anomaly detection methods
$T_{ML} \leftarrow$ Threshold number of voting for machine learning models
PreProcessing on the input dataset D which is described in Algorithm 1
Store the result of anomaly detection methods of AD SET on D_1 into $U_1, \cdots, U_N$ (0 for normal & 1 for anomaly).
for Each row of D_1 **do**
    Set $U_{Total} = \sum_{i=1}^{N} U_i$
    if $U_{Total} \geq T_{AD}$ **then**
        Add this row into the set Y_1
    else
        Label the row as Potential Normal
    end if
end for
for Each row of Y_1 **do**
    Store the result of EAD method into C_1
    if $C_1 = 1$ **then**
        Label the row as Potential Fraud
    else
        Label the row as Potential Normal
    end if
end for
Create F_1 dataset as the subset of Y_1 for which labeled as Potential Fraud.
Send F_1 dataset for experts.
Receive the opinion of experts on F_1 dataset as E_1
Create E_1' as oversampling SMOTE model of $E_1 \cup G_1$ where G_1 is a random subset of size PN% of D_1 for which labeled as Potential Normal
for each Model of ML SET **do**
    Learn the Model with E_1' Dataset as training set.
end for
Set trained models as TS_1
for j from 2 to m **do**
    for each row of D_j **do**
        Use TS_{j-1} for prediction on D_j as tast set.
        Store the response prediction of TS_{j-1} on D_j into $V_1, \cdots, V_M$ (0 for normal & 1 for anomaly).
        Set $V_{Total} = \sum_{i=1}^{M} V_i$
        if $V_{Total} \geq T_{ML}$ **then**
            Label the row as Potential Fraud
        else
            Label the row as Potential Normal
        end if
        Create F_j dataset as the subset of D_j for which labeled as Potential Fraud.
        Send F_j dataset for experts.
        Receive the opinion of experts on F_j dataset as E_j
        Create E_j' as oversampling SMOTE model of $E_j \cup G_j \cup E_{j-1}'$ where G_j is a random subset of size PN% of D_j for which labeled as Potential Normal
        for each Models of ML SET **do**
            Learn the Model with E_j' Dataset as training set.
        end for
        Set trained models as TS_j
    end for
end for

3.4 EVALUATION

At each stage of the algorithm, we can validate the models after receiving expert opinions. Different criteria are used for evaluating algorithms, and here, we have used three criteria: accuracy, recall, and F1, which we will introduce below.

Accuracy/Weighted Recall metric: Accuracy measures the ratio of correctly predicted instances to the total number of instances. It is calculated

as $Acc = \frac{TP+TN}{TP+TN+FP+FN}$, where TP and TN represent true positives and true negatives (correct predictions), while FP and FN denote false positives and false negatives (incorrect predictions), respectively [Abakarim et al., 2023].

F1-score metric: F1-score represents the harmonic mean of precision and recall. It is calculated as $F1 = \frac{2 \times PS \times RS}{PS+RS}$, providing a balanced measure of both precision and recall [Abakarim et al., 2023]. Higher F1-scores indicate better model performance, considering both false positives and false negatives.

Recall/Sensitivity metric: Recall measures the ratio of correctly predicted positive instances to the total number of positive instances. It is calculated as $RS \text{ or } Recall \ Score = \frac{TP}{TP+FN}$, where TP and FN represent true positives and false negatives, respectively [Abakarim et al., 2023]. Higher recall indicates a better ability to identify positive instances.

4 Real Data Application

Now, we demonstrate the practical application of our potential anomaly detection framework using real-world datasets. This involves integrating the anomaly detection methods, machine learning techniques, and evaluation metrics discussed earlier to create a comprehensive approach for identifying fraudulent transactions. The implementation showcases the effectiveness of our method in handling imbalanced datasets and adapting to evolving data patterns, ultimately enhancing the accuracy and reliability of fraud detection in various application domains.

Carclaims dataset

The **Auto Insurance Claims Fraud Detection** dataset on Kaggle, compiled by Anto Pravin John Bosco, contains records to identify fraudulent insurance claims. It includes various features related to automobile insurance policies and claims, allowing the application of machine-learning techniques for anomaly detection. The dataset, comprising 15,420 observations and 31 features, serves as a valuable resource for developing fraud detection models in the insurance industry. Note that our dataset includes the binary variable of cheating [Bosco, 2020].

Table 1: Frequency table for some of categorical variables

Feature	Category	Percent of Category	Frequency
Sex	Female	15.69	2420
	Male	84.31	13000
MaritalStatus	Divorced	0.49	76
	Married	68.90	10625
	Single	30.38	4684
	Widow	0.23	35
VehicleCategory	Sedan	62.72	9671
	Sport	34.75	5358
	Utility	2.54	391
DriverRating	1	25.58	3944
	2	24.65	3801
	3	25.19	3884
	4	24.58	3791
FraudFound	No	94.01	14497
	Yes	5.99	923

Algorithm Specifications

In this research, by implementing the structures and models as mentioned below, we reached the following results.

1. Preprocessing Algorithm:
   For all scenarios executed in this algorithm, the following values have been considered:

$$m = 4, p_1 = 50\%, p_2 = 30\%, p_3 = 10\%, p_4 = 10\%$$

2. Simple Structure A:
   This is the Simple structure of the SAD algorithm where the set of machine learning models and outlier detection methods are as follows:
   $PN = 10\%$,
   AD SET $\leftarrow$ (Box plot, Percentile, Z-Score, Hample filter, E-Grubs test, LOF, KNN, Isolation forest, KModes),
   ML SET $\leftarrow$ (LR, SVM, XGBoost, KNN, RF, Adaboost, NN).

3. Simple Structure B:
   This is the Simple structure of the SAD algorithm where the set of machine learning models and outlier detection methods are as follows:
   $PN = 10\%$,
   AD SET $\leftarrow$ (Box plot, Percentile, Z-Score, Hample filter, E-Grubs test, LOF, KNN, Isolation forest, KModes, EAD),
   ML SET $\leftarrow$ (LR, SVM, XGBoost, KNN, RF, Adaboost, NN).

4. Multilayered structure of the SAD algorithm:
   The set of machine learning models and outlier detection methods for this structure are as follows:
   $PN = 10\%$,
   AD SET $\leftarrow$ (Box plot, Percentile, Z-Score, Hample filter, E-Grubs test, LOF, KNN, Isolation forest, KModes, EAD),
   ML SET $\leftarrow$ (LR, SVM, XGBoost, KNN, RF, Adaboost, NN).

Table of Evaluation Metrics

Table 2: Evaluation metrics of different algorithms along with the proposed algorithm by different partitions and structures

		Simple A			Simple B			Multilayer		
ML Model	Partition	Accuracy	F1	Recall	Accuracy	F1	Recall	Accuracy	F1	Recall
Adaboost	2	0.72	0.83	0.75	0.74	0.85	0.77	0.82	0.90	0.86
	3	0.69	0.81	0.72	0.72	0.83	0.75	0.81	0.89	0.84
	4	0.72	0.83	0.75	0.71	0.83	0.74	0.84	0.91	0.88
KNN	2	0.71	0.82	0.73	0.69	0.81	0.72	0.73	0.84	0.75
	3	0.68	0.81	0.71	0.69	0.82	0.72	0.77	0.86	0.80
	4	0.72	0.84	0.75	0.75	0.85	0.78	0.82	0.90	0.86
ML	2	0.80	0.89	0.84	0.77	0.87	0.81	0.84	0.91	0.88
	3	0.76	0.86	0.80	0.80	0.89	0.84	0.80	0.88	0.83
	4	0.78	0.87	0.81	0.80	0.89	0.83	0.81	0.89	0.84
NB	2	0.83	0.90	0.87	0.83	0.91	0.88	0.85	0.92	0.89
	3	0.81	0.90	0.85	0.81	0.90	0.85	0.85	0.92	0.88
	4	0.84	0.91	0.88	0.83	0.91	0.87	0.85	0.92	0.88
ANN	2	0.81	0.89	0.85	0.78	0.87	0.81	0.83	0.91	0.87
	3	0.78	0.88	0.82	0.81	0.89	0.85	0.87	0.93	0.91
	4	0.79	0.88	0.82	0.80	0.89	0.84	0.85	0.92	0.89
RF	2	0.87	0.93	0.92	0.86	0.93	0.92	0.93	0.96	0.99
	3	0.89	0.94	0.94	0.89	0.94	0.94	0.94	0.97	0.99
	4	**0.91**	0.95	0.96	**0.92**	0.96	0.97	**0.95**	0.97	1.00
SVM	2	0.80	0.89	0.84	0.78	0.87	0.82	0.78	0.88	0.85
	3	0.74	0.85	0.78	0.79	0.88	0.83	0.72	0.84	0.75
	4	0.78	0.87	0.81	0.80	0.89	0.83	0.77	0.87	0.80
XGBoost	2	0.82	0.90	0.87	0.79	0.88	0.83	0.71	0.83	0.74
	3	0.84	0.91	0.88	0.85	0.92	0.90	0.78	0.87	0.80
	4	0.83	0.91	0.87	0.86	0.92	0.91	0.71	0.83	0.74
SAD	2	**0.88**	0.94	0.93	**0.89**	0.94	0.95	**0.93**	0.96	0.99
	3	**0.89**	0.94	0.94	**0.91**	0.95	0.96	**0.94**	0.97	0.99
	4	**0.94**	**0.97**	0.99	**0.94**	**0.97**	**0.99**	**0.95**	**0.97**	**1.00**

As you can see in Table 4, the SAD model has performed very well compared to other models. The models were examined in terms of accuracy, F1 and Recall in three simple A, simple B and Multilayered structure of the SAD algorithm. After the SAD model, the random forest model had

the highest performance. As the cycle and process progresses, that is, from partition 2 onwards, we see an increase in accuracy, etc., in the majority of models, because as we move forward, the model learns more and becomes stronger. From the point of view of the structure, the multi-layer structure has performed better, and according to table 4, it has had a much shorter duration than the other two structures.For example, the highest performance in all cases is for the fourth partition of the SAD model and the Multilayered structure of the SAD algorithm, where the accuracy, F1, and recall are 95%, 97%, and 100%, respectively.

Table 3: Evaluation metric of SAD algorithm for different threshold values and different structures

		Simple A			Simple B			MultiLayer		
T_{ML}	T_{AD}	F1	Recall	Accuracy	F1	Recall	Accuracy	F1	Recall	Accuracy
	1	0.54	0.38	**0.39**	0.54	0.38	0.39	0.78	0.67	0.65
	2	0.57	0.41	**0.42**	0.60	0.44	0.45	0.79	0.67	0.65
3	3	0.77	0.64	**0.63**	0.77	0.54	0.54	0.79	0.67	0.66
	4	0.84	0.75	**0.72**	0.69	0.78	0.75	0.84	0.76	0.73
	5	0.88	0.82	**0.79**	0.85	0.79	**0.76**	0.84	0.76	**0.73**
	1	0.72	0.58	0.57	0.68	0.54	0.53	0.87	0.81	0.78
	2	0.75	0.62	0.61	0.72	0.58	0.57	0.87	0.80	0.77
4	3	0.88	0.82	0.79	0.80	0.69	0.67	0.92	0.89	0.85
	4	0.91	0.87	0.83	0.90	0.86	0.82	0.92	0.89	0.84
	5	0.92	0.90	**0.86**	0.92	0.89	**0.85**	0.92	0.90	**0.85**
	1	0.82	0.73	0.70	0.80	0.69	0.67	0.93	0.91	0.87
	2	0.81	0.71	0.68	0.82	0.73	0.70	0.94	0.92	0.88
5	3	0.91	0.88	0.83	0.88	0.82	0.79	0.95	0.96	0.91
	4	0.94	0.93	0.88	0.93	0.92	0.88	0.95	0.95	0.90
	5	0.94	0.94	**0.89**	0.94	0.94	**0.89**	0.96	0.98	**0.93**
	1	0.88	0.82	0.78	0.87	0.81	0.78	0.96	0.97	0.92
	2	0.87	0.80	0.77	0.88	0.82	0.78	0.96	0.97	0.92
6	3	0.94	0.94	0.90	0.93	0.92	0.87	0.97	0.99	0.94
	4	0.95	0.96	0.91	0.95	0.96	0.91	0.97	1.00	0.94
	5	0.96	0.98	**0.92**	0.96	0.97	**0.92**	0.97	1.00	**0.95**
	1	0.94	0.95	0.90	0.94	0.92	0.88	0.97	0.99	0.94
	2	0.94	0.93	0.88	0.93	0.91	0.87	0.97	1.00	0.94
7	3	0.96	0.97	0.92	0.96	0.96	0.92	0.97	1.00	0.94
	4	0.97	0.99	0.94	0.97	0.97	0.94	0.97	1.00	0.95
	5	0.97	0.99	**0.94**	0.97	0.97	**0.94**	0.97	1.00	**0.94**

As demonstrated in Table 3, when T_{ML} increases while T_{AD} remains constant, we observe an improvement in accuracy, recall, and F1-score. For instance, when T_{ML} equals 3 and T_{AD} is set to 5, the accuracy in the simple structure A rises from 79% to 95%, and in the Multilayered structure of the SAD algorithm, it increases from 73% to 94%. This indicates a direct relationship between increasing T_{ML} and enhancing accuracy.

Similarly, with a constant T_{ML} and an increasing T_{AD}, we notice a growth in accuracy. In simple structure A, when T_{ML} is 3 and T_{AD} increases from 1 to 5, accuracy also increases from 39% to 79%, demonstrating a direct relationship between the two parameters.

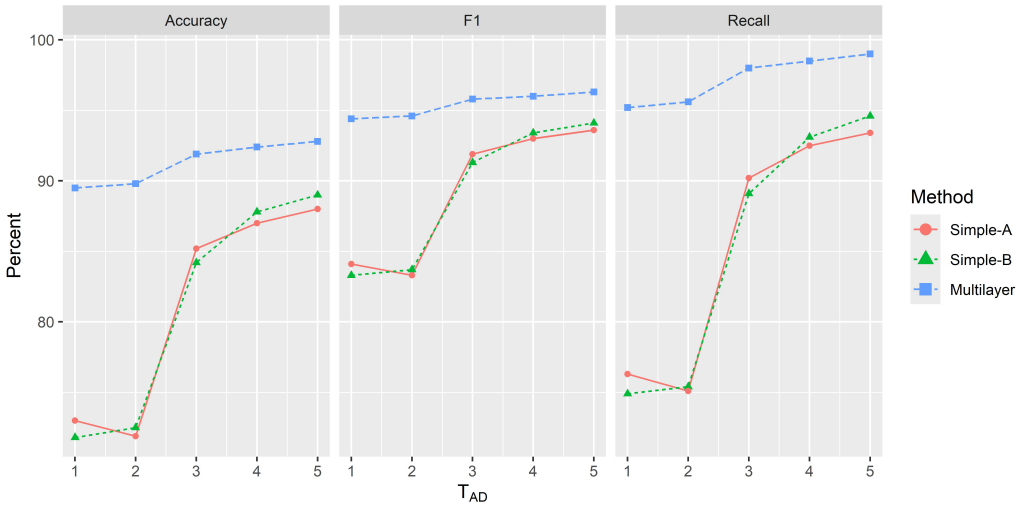


Figure 3: The trend of evaluation metric of SAD algorithm respect to T_{AD} in the second partition for $T_{ML} = 7\,, M = 8$ and different structures

In Figure 3, with T_{ML} fixed at 7 for the first partition, it becomes evident that accuracy exhibits a positive correlation with T_{AD}. Specifically, in Simple-A and Simple-B structures, an increase in T_{AD} from 2 to 3 yields a significant upward jump in accuracy. However, the Multilayered structure of the SAD algorithm generally outperforms the other two structures. This trend of the multilayer structure's superior performance can also be observed in the F1-score and Recall metrics.

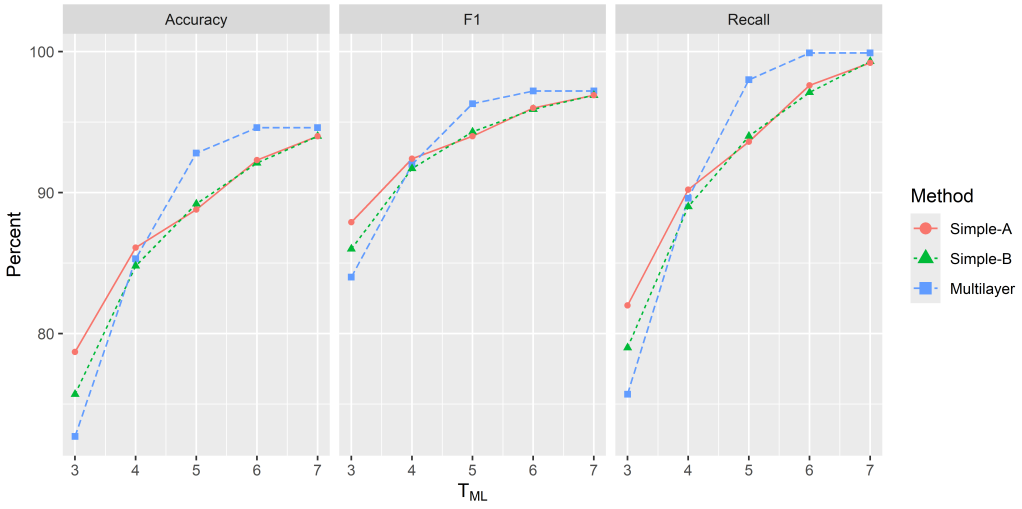


Figure 4: The trend of evaluation metric of SAD algorithm respect to T_{ML} in the forth partition for $T_{AD} = 5\,, M = 8$ and different structures

In Figure 4, with T_{AD} set to 5 for the third partition, we observe that accuracy demonstrates a positive correlation with T_{ML}. In Simple-A and Simple-B structures, an increase in T_{ML} from 3 to 5 results in a noticeable upward jump in accuracy. However, the Multilayered structure of the SAD algorithm consistently outperforms the other two structures. Similar trends can be observed for the F1-score and Recall metrics. Notably, in the multilayer structure, the rate of change appears to stabilize when T_{ML} reaches 6 or higher.

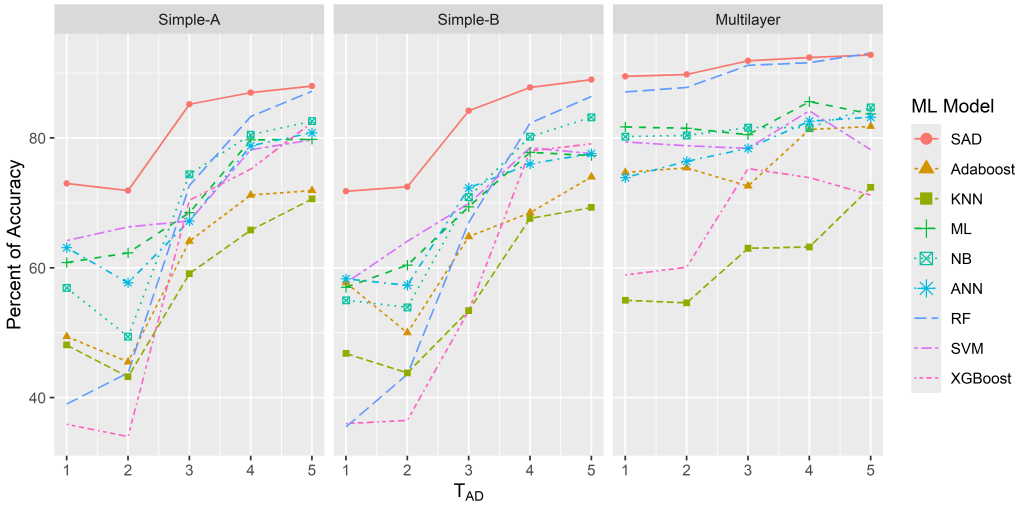


Figure 5: The trend of accuracy of algorithms respect to T_{AD} in the second partition for $T_{ML} = 7\,, M = 8$ and different structures.

In Figure 5, with T_{ML} fixed at 7 for the first partition, the accuracy of various machine learning models within different structures is depicted. It becomes evident that accuracy shows a positive correlation with T_{AD} across all machine learning models. This suggests that increasing the anomaly detection threshold generally improves the performance of these models, regardless of the specific structure in which they operate.

In general, the Multilayered structure of the SAD algorithm consistently outperforms the other two structures, demonstrating its superior performance. Furthermore, among the various models analyzed, the SAD model emerges as the top-performing model across all structures. This highlights the effectiveness of the SAD model in anomaly detection tasks within different structural contexts.

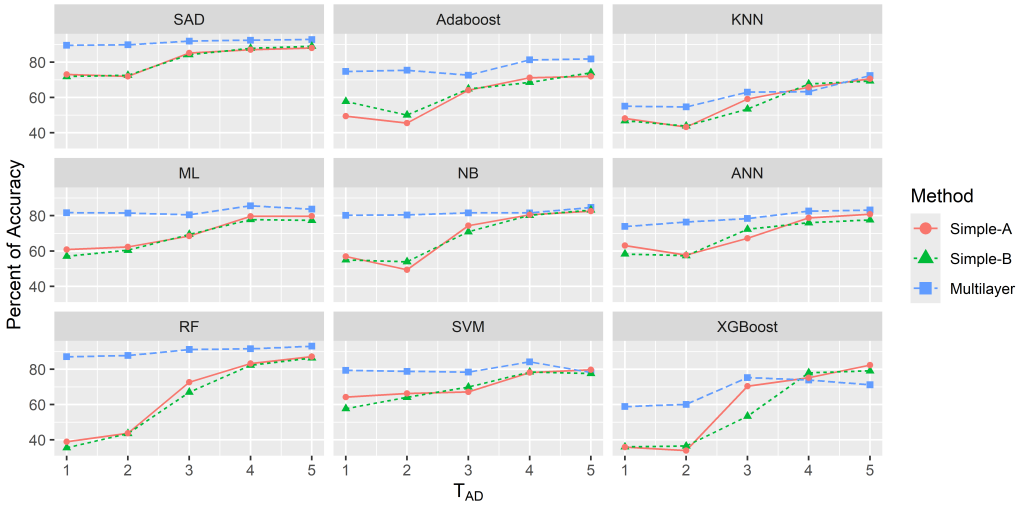


Figure 6: The trend of accuracy of structures respect to T_{ML} in the forth partition for $T_{AD} = 5\,, M = 8$ and different algorithms

IIn Figure 6, with T_{ML} set to 7 for the first partition, the accuracy of various machine learning models within different structures is presented. We observe that accuracy generally exhibits a positive correlation with T_{AD} across all models. However, this correlation is less pronounced in the SAD model, as its performance improvement is more closely tied to increases in T_{ML}. Overall, the Multilayered structure of the SAD algorithm maintains its superior performance compared to the other two structures, while the SAD model continues to stand out among the other models in terms of accuracy.

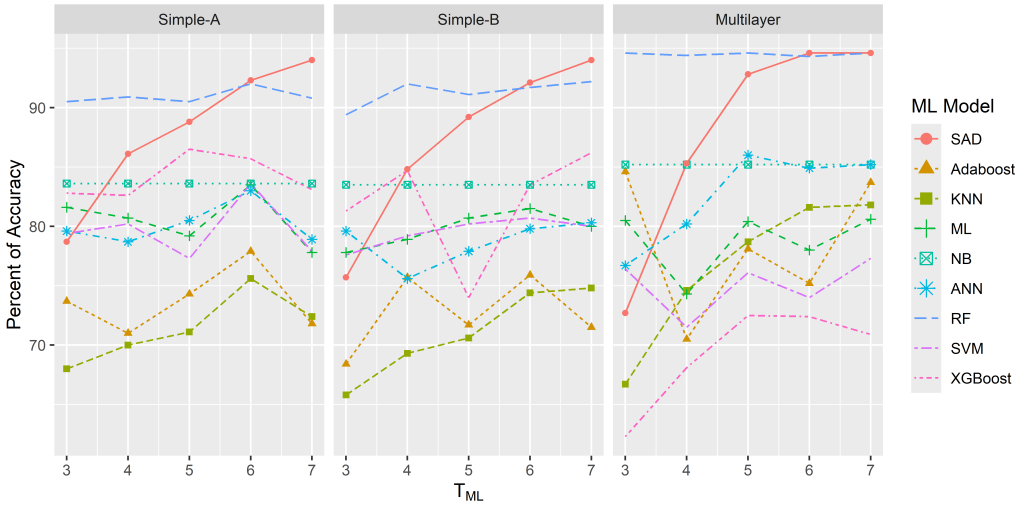


Figure 7: The trend of accuracy of algorithms respect to T_{ML} in the forth partition for $T_{AD} = 5\,, M = 8$ and different structures.

In Figure 7, with T_{AD} set to 5 for the third partition, we analyze the accuracy of various machine learning models within different structures. The SAD model displays a strong correlation between accuracy and T_{ML}, while other machine learning models exhibit a weaker correlation or fluctuations. Ultimately, the SAD model achieves higher accuracy across all three structures, underscoring its superior performance. Additionally, Random Forest and Logistic Regression models also show some degree of independence from the correlation, further highlighting the diverse performance characteristics of these models within different structural contexts.

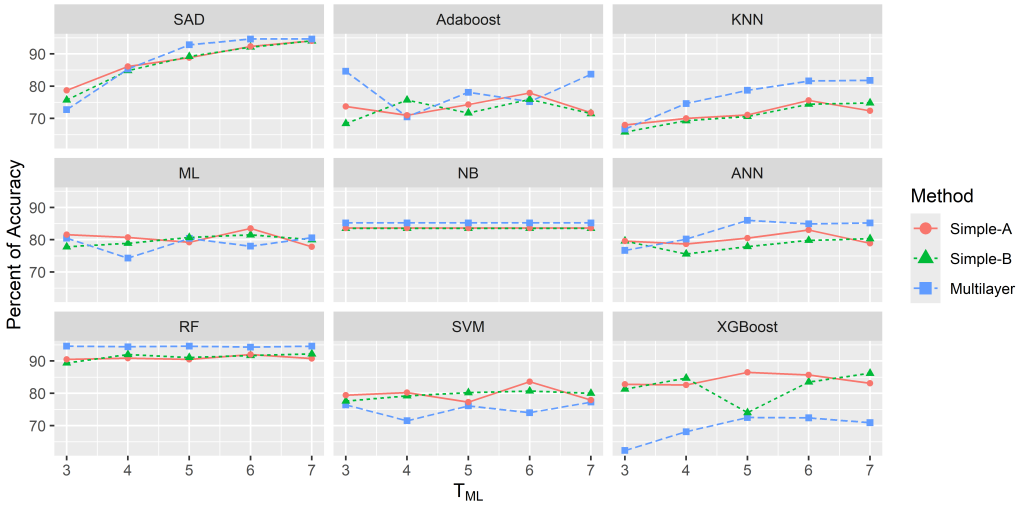


Figure 8: The trend of accuracy of structures respect to T_{ML} in the forth partition for $T_{AD} = 5\,, M = 8$ and different algorithms

In Figure 8, with T_{AD} fixed at 5 for the third partition, the accuracy of various machine learning models within different structures is depicted. The SAD model shows a positive correlation between accuracy and T_{ML}, whereas other machine learning models display a weaker correlation. This indicates that the performance improvement in the SAD model is more directly connected to increases in T_{ML} compared to other models, which rely more on T_{AD} adjustments for performance enhancements.

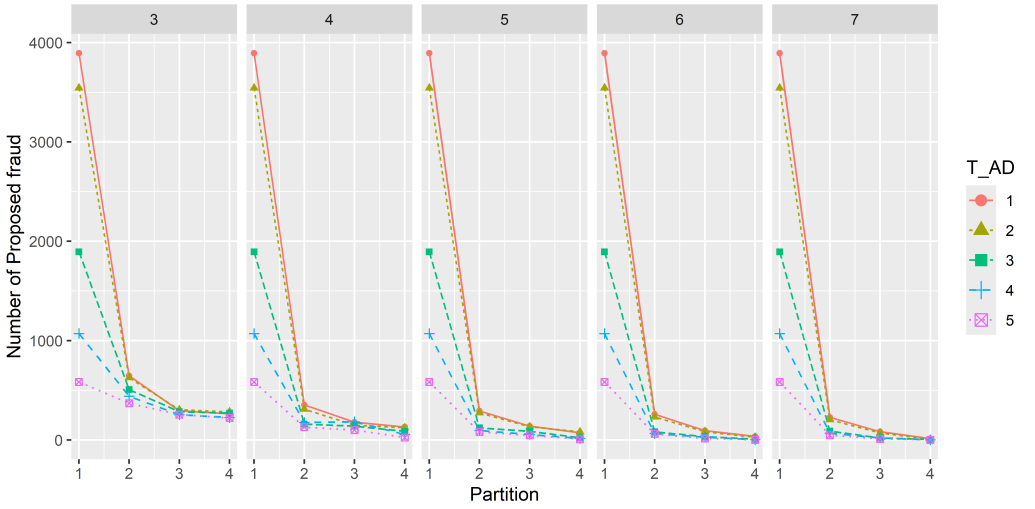


Figure 9: Number of potential frauds for different values of T_{AD} , T_{ML} respect to partition number.

As illustrated in Figure 9, for data sections ranging from 1 to 4 with varying T_{ML} values, a higher T_{AD} results in fewer data points being sent to the insurance company for further examination. As the system evolves and T_{ML} progressively increases across sections, the number of data points requiring expert review significantly decreases, indicating the improved efficiency and accuracy of the anomaly detection system.

For instance, with $T_{AD} = 5$ and $M = 8$ as parameter values, the system can achieve high accuracy, comparable to that of the SAD method shown in the figures and tables above, by submitting merely 10% (PN) of the data to the insurance company. This result demonstrates a significant improvement in efficiency compared to manually reviewing each case, ultimately saving time and resources for the organization.

Table 4: Elapsed times in each of the partitions with the SAD algorithm for $T_{ML} = 7\,, T_{AD} = 5\,, M = 8$.

Partition	Structure	Time (minutes)
1	Simple-A	0.53
	Simple-B	25.18
	Multilayer	1.78
2	Simple-A	0.78
	Simple-B	1.03
	Multilayer	0.80
3	Simple-A	0.06
	Simple-B	0.07
	Multilayer	0.06
4	Simple-A	0.07
	Simple-B	0.09
	Multilayer	0.07
Total	Simple-A	1.43
	Simple-B	26.38
	Multilayer	2.71

Table 4 shows that the first partition requires more time due to higher data volume and entropy structure implementation. Simple Structure A is faster due to the absence of the entropy method, while Simple Structure B takes longer despite using entropy. The Multilayered structure of the SAD algorithm significantly reduces time from 25.18 minutes to 1.78 minutes with an entropy filter layer. In partitions 2 to 4, no significant differences are observed between models due to the same $ML\,SET$ implementation. The upcoming section provides a conclusion and recommendations for further exploration in this field.

5 CONCLUSION

This study leveraged artificial intelligence and reinforcement learning to detect insurance fraud. Various algorithms and structures, including statistical methods and machine learning models, were presented, leading to the following key findings: (**1**) Due to time constraints, the entropy method and information theory are unsuitable for the Simple-B structure, as all observations in the first data section must be reviewed by the entire $AD\,SET$; (**2**) Higher T_{AD} and T_{ML} parameters improve accuracy across all structures, with more methods employed in the initial voting process; (**3**) Increasing T_{AD} reduces the number of observations sent for expert review, while raising T_{ML} decreases the amount of data sent from the third section onwards; (**4**) The SAD method provides the best accuracy within the dataset, followed by the

Random Forest model; (**5**) Increasing T_{AD} significantly improves accuracy in the second section, while increasing T_{ML} primarily benefits the SAD method; (**6**) The same observations on accuracy also apply to recall and F1 score.

Recommendations for future research include optimizing utilized models, exploring different $ML\,SET$s, employing various $AD\,SET$s, and e xamining diverse logistic regression models. Weighted voting may also enhance the structures' effectiveness, allowing more established models and methods to have a greater influence in the voting process.

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